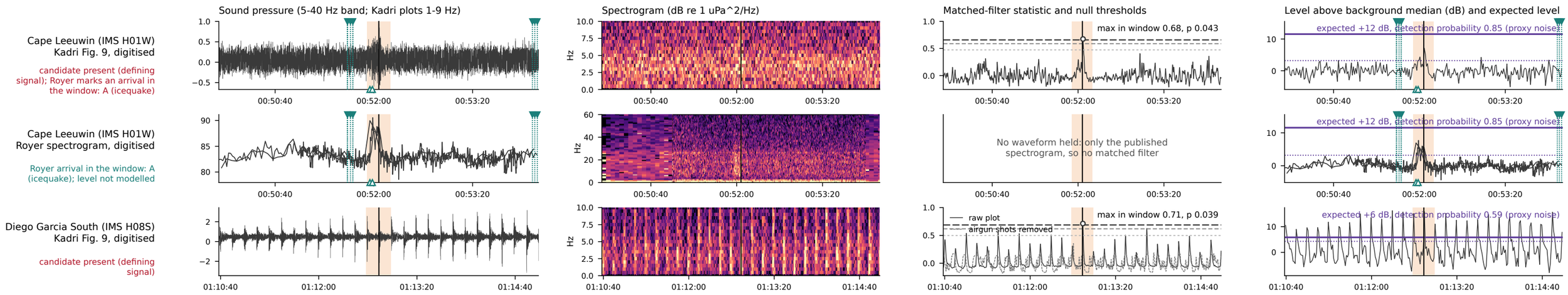
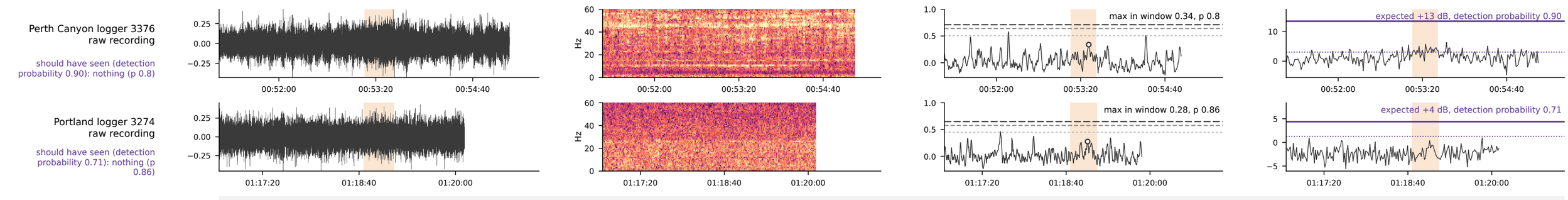
If the Cape Leeuwin 00:52:04 and Diego Garcia South 01:12:49 signals were one impact (33.6 S 94.5 E, impact 00:31:41 UTC): the data held at every receiver when its sound should arrive

needed coupling 0.068 (326 times the F-35A value 2.1e-4; P_D = 0.5 jointly at Cape Leeuwin and Diego Garcia South).
Bearing check: the source point is at 268.9 deg from Cape Leeuwin (1,812 km) and 144.8 deg from Diego Garcia South (3,660 km). Kadri gives 57 deg or 306.18 deg for the 00:52 signal (text p. 9 and Table 1 caption disagree); neither is within 37 deg of 268.9 deg. The nearest Table 1 bearing is 00:52:50 at 274.84 deg, 46 s later.



Diego Garcia North, (IMS H08N) — No recording held. Predicted arrival 01:15:02 to 01:15:22 UTC. no recording obtained (IMS data not released to the project).

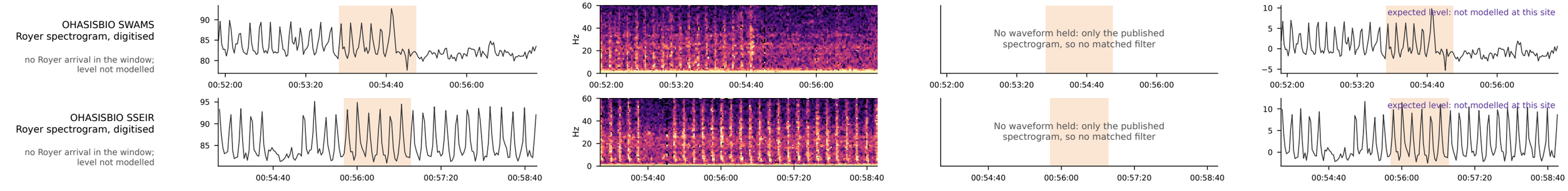
Perth Canyon logger 3315, raw recording — No recording held. Predicted arrival 00:53:10 to 00:53:34 UTC. no record overlaps the window (records are 307 s every 15 min, alternating slots).



Portland logger 3275, raw recording — No recording held. Predicted arrival 01:18:42 to 01:19:09 UTC. no record overlaps the window (records are 307 s every 15 min).

Scott Reef logger 3250, raw recording — No recording held. Predicted arrival 01:09:00 to 01:09:44 UTC. no record overlaps the window (records are 307 s every 15 min).

OHASISBIO NEAMS — No recording held. Predicted arrival 00:43:23 to 00:44:26 UTC. instrument lost on recovery: no data exist (Royer 2016 Table 1).



OHASISBIO MAD, Royer spectrogram, digitised — No recording held. Predicted arrival 01:11:42 to 01:12:31 UTC. no digitised Royer panel covers the window; raw data not held.

OHASISBIO WKER, Royer spectrogram, digitised — No recording held. Predicted arrival 01:07:10 to 01:08:18 UTC. no digitised Royer panel covers the window; raw data not held.

OHASISBIO NCRO, Royer spectrogram, digitised — No recording held. Predicted arrival 01:13:07 to 01:14:09 UTC. no digitised Royer panel covers the window; raw data not held.

Time (UTC, 8 March 2014). Orange band: predicted arrival window (95 %). Black line: the signal that defines the candidate. Purple line: expected level of the candidate if it were the impact; dotted purple: level for detection probability 0.5.
Grey dashed lines: matched-filter thresholds for p = 0.3, 0.1, 0.05 (light to dark). Teal: Royer's picked arrivals (filled triangle and dotted line: his Table 5; open triangle: the arrival marks on his figures).

(i) Each row is one receiver. The orange band shows when sound from the candidate source would arrive at that receiver. A grey row means that we hold no recording for that time. The purple line shows how loud the candidate would be if it were the impact, with the coupling that it needs. If the purple line is high and the trace shows nothing above the dotted purple line, the receiver should have detected the sound and did not. Plots marked digitised are copies of published figures, not recordings. These are model predictions. The sound-loss model and the coupling are not calibrated.

(ii) Source: lag locus on the 7th-arc grid (offset 0), Monte Carlo n = 4000, c ~ N(1.482, 0.006) km/s, pick error sd 3.6 s (injection 3 s + clock 2 s). Window = 2.5-97.5 % of predicted arrival incl. c and pick errors; plots window +/- 2 min. Level: E 1.08 GJ, tau90 3.3 s (composition coincidence_locus.csv (v2)); SNR = peak 1 s 5-40 Hz level re 14-day background median, TL from the arc grid (IMOS: RAM-corrected table imos_tl_ram_arc.csv; IMS: KRAKEN - RAM delta, +4.8 dB triad gain), P_D = block 7 logistic fit, M1b, alpha 0.05 (IMS: proxy noise). M1b = envelope matched filter (18-template KRAKEN bank on IMOS, station templates on panels); p = (k+1)/(n+1) against 55 s window maxima of the logger's own 1-15 March background (3315 <- 3376, 3275 <- 3274 stand-ins) or, for panels, IMOS 3376 through the plot emulator; p for the window maximum is not corrected for window length. Kadri Fig. 9 panels: digitised plots, 20 Hz, nothing above ~7 Hz trusted; Kadri times: text time 00:52:03, caption time 00:54:30. Royer (2016) spectrograms: colour->dB inversion (+/-1.5 dB), band 5-40 Hz; picks from Table 5 (H01W figure labels run 13.6-16 s later than the table, unresolved). PREDICTIVE, NOT CALIBRATED; caveats C1 (slope misfit +10.4/+7.4 dB/octave at H01W/H08S) and C2 (coupling not calibrated; F-35A anchor 2.1e-4).